📄 **PROJECT REPORT**

Smart Waste Segregation Prediction

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1. Introduction

Rapid urbanization has increased waste generation, making proper waste segregation a major challenge. Manual segregation is time-consuming, error-prone, and inefficient. To address this, we developed an AI-based Waste Classification System that automatically identifies waste type from images using deep learning.

2. Problem Statement

- Manual waste sorting is:
- Time-consuming ⏳
- Prone to human error ❌
- Inefficient for large-scale recycling ♻️

There is a need for an automated system that can accurately classify waste into categories such as recyclable, organic, metal, glass, etc.

3. Proposed Solution

We built a deep learning-based image classification system that:

- Accepts waste images as input 📷
- Processes images using a trained CNN model 🧠
- Predicts waste category instantly ⚡
- Helps in proper segregation for recycling ♻️

4. System Architecture

Flow of the system:

- User uploads image
- Image preprocessing (resize, normalize)

- CNN model prediction
- Output displayed (waste category)

5. Architecture Diagram

User Upload Image

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Streamlit Web App

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Image Preprocessing (224x224, normalization)

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CNN Deep Learning Model

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Prediction Output (Plastic / Metal / Glass / Organic etc.)

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Display Result to User

6. Technology Stack

- Python 🐍
- TensorFlow / Keras 🧠
- Streamlit 🌐
- NumPy
- Pillow (PIL)
- Git & GitHub

7. Machine Learning Model

We used a Convolutional Neural Network (CNN) because:

- It is highly effective for image classification
- It extracts spatial features automatically
- Provides good accuracy for visual datasets

8. Dataset Description

The dataset consists of images categorized into:

- Plastic
- Metal
- Glass
- Paper
- Organic Wast

Images were resized and normalized before training.

9. Implementation

Steps followed:

- Data collection and preprocessing
- Model design using CNN layers
- Model training and validation
- Saving trained model
- Integration with Streamlit web app
- Deployment on cloud platform

10. Results

- The model successfully classifies waste images
- Provides real-time predictions
- Helps automate waste segregation process

11. Applications

- Smart cities 🏙️
- Waste management systems ♻️
- Recycling industries
- Schools & awareness systems
- Municipal corporations
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12. Future Improvements

- Increase dataset size for better accuracy
- Use advanced models like EfficientNet
- Add real-time camera detection

13. Conclusion

This project successfully demonstrates how Artificial Intelligence can improve waste management systems. By automating waste classification, it reduces human effort and supports sustainable recycling practices.

14.Live Deployment

The model has been deployed using cloud hosting, and the application is accessible via the following live URL:

https://smart-waste-classifi-9pwq.bolt.host

15. GitHub Repository

The source code of the project is available at:

https://github.com/saniyashariff61281-sys/SMART-WASTE-CLASSIFIER.git